CSE341: Microprocessors

A. Course General Information:

Course Code:	CSE 341
Course Title:	Microprocessors
Credit Hours (Theory + Lab):	3+0
Contact Hours (Theory + Lab):	3+3
Category:	Program Core
Type:	Required, Engineering, Lecture + Laboratory
Prerequisites:	CSE 260: Digital Logic Design

B. Course Catalog Description:

This course introduces the architecture, working mechanism and functionality of different microprocessors. This course primarily focuses on the Intel 8086 microprocessor's architecture, its memory partitioning and registers, the instruction set, timing diagrams and its interrupt structure. Additionally, the concepts I/O addressing and the idea behind Direct Memory Access and its cascading are also discussed. Lastly the course concludes with some concepts of the more advanced Intel 80x86 microprocessors and their basic architecture. The course includes a compulsory 3-hour laboratory work each week.

C. Course Objective

The objectives of this course are to:

a. Getting familiar with the components of a microcomputer system and getting associated with the internal architecture of Intel 8086 microprocessor.

b. Introducing the instruction set of 8086 microprocessor, its different addressing modes, and address translation techniques.

c. Discuss the concepts of clock signals and formulate timing diagram.

d. Explain interrupt handling in 8086 and the basic functionalities of the 8259A PIC

e. Assess the advantages of direct memory access and discuss the 8237 DMA controller.

f. Introducing later processors (80x86) and compare their components and basic features.

g. Understand assembly language, develop assembly language programs and learn how to interface them to various devices using the hardware kit.

D. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	**Explain** the basic components of a microcomputer system, and the Internal Architecture of the Intel 8086 along with its memory partitioning and address translation techniques.	25
CO2	**Illustrate** the instruction set and addressing modes of the 8086 microprocessor, its clock signals and timing diagrams and machine code generation.	20
CO3	**Illustrate** the concepts of interrupt handling in the 8086, the working principle behind the 8259A PIC, I/O addressing and the inclusion of Direct Memory Access for the Intel 8086.	20
CO4	**Assess** the features and components of 80x86 processors.	15
CO5	**Create** programs using assembly language and learn how to interface them to various devices using the hardware kit.	20

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods an d activities	Assessment tools
CO1	**Explain** the basic components of a microcomputer system, and the internal architecture of Intel 8086 microprocessor, its memory segmentation and partitioning mechanism, and the concept of address translation using physical addressing	**a**	Cognitive	Lectures, Discussion, Q/A	Quiz, Assignment Midterm,
CO2	**Illustrate** the different addressing modes and instructions supported by the 8086 microprocessor, the ideas behind timing diagrams, memory banking and machine code generation.	**a**	Cognitive	Lectures, Discussion, Q/A	Quiz, Assignment Midterm, Final
CO3	**Illustrate** how the Intel 8086 handles incoming interrupts the working mechanism of the 8259A PIC, I/O addressing and the inclusion of Direct Memory Access for the Intel 8086.	**a**	Cognitive	Lectures, Discussion, Q/A	Quiz, Assignment Final,

CO4	**Assess** and compare the different features and components of the Intel 80186 and 80286 processors.	**a**	Cognitive	Lectures, Discussion, Q/A	Quiz, Assignment Final,
CO5	**Create** assembly language programs and learn how to interface them to various devices using the hardware kit.	**i**	Psychomotor, Affective	Lectures, Discussion, Q/A, Lab Tasks	Lab works, project & viva

F. Course Materials:

i. Text and Reference Books:

Sl.	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1.	Microprocessors and Interfacing	Douglas V. Hall	1992	2nd	McGraw Hill	ISBN: 978-0071004626
2.	Assembly Language Programming Organization of the IBM PC	Ytha Yu, Charles Marut	1992	International Ed.	McGraw Hill	ISBN: 978-0070726925
3.	Intel Microprocessors, Architecture, Programming and Interfacing	Barry B. Brey	2009	8th	Pearson Prentice Hall	ISBN–13: 978–0–13–502645–8

ii. Other materials (if any)

a. Lecture notes and presentation slides
b. Lab hand-outs
c. Lab usage manual
d. Emulator (emu8086)

H. **Lesson Plan (Theory):**

No	Topic	Week	Related CO (if any)
1.	Introduction to microprocessors, Basics of microcomputer and microcontroller systems, Memory partitioning and	1,2,3	CO1

	segmentation, address translation and physical addressing in the Intel 8086		
2.	Internal architecture of Intel 8086, Introduction to the 8086 flag register, Hardware specification, Description of minimum and maximum mode pins	4,5	CO1
3.	Addressing Modes, Machine Code, Clock signals and Timing diagrams, Memory Banking	6,7,8	CO2
4.	Software & Hardware interrupts, Interrupt vectors and interrupt vector table, Interrupt handling, 8259A Programmable Interrupt Controller (PIC), cascading of 8259As	9,10	CO3
5.	Addressing I/O devices, Basic I/O System, DMA, 8237 DMA Controller	11,12	CO3
6.	Features of 80186 and 80286 processors, internal block diagram and functional parts of these processors, introduction to pipelining in their architecture.	13,14	CO4

Lesson Plan (Lab):

No	Topic	Week	Related CO (if any)
1.	Introduction to the basics of a microprocessor and intel8086 chipset, along with the emu8086, the simulator used to code in assembly language, representation of numbers and characters in assembly language.	**1,2**	**C05**
2.	Explaining the basic idea on the registers involved in 8086 processor, how the registers can be used as variables and special features of some register, performing basic arithmetic operations.	**3**	
3.	Introduction to Interrupts for basic input output mechanisms.	**4**	
4.	Familiarize the students with how decisions can be made with the jump instruction, conditional and unconditional jump, implement high-level language decision structures using assembly.	**5**	
5.	Introduction to loop instructions, different looping structures.	**6**	
6.	Make students acquainted with how one-dimensional arrays are declared in assembly language along with the operations they are used in and how array elements can be accessed by using different addressing modes.	**7**	
7.	Learning basic push pop operations using stack and solving different problems in more effective way using stack.	**8**	
8.	Introduce the students to get the basic idea of Macros and Procedures and understand how they work.	**9, 10**	

9.	Introduction to the basics of an EMU/MDA8086 Trainer Kit and its components. Hands on knowledge on how to configure the kit using "Machine Code" mode.	**11, 12**
10.	Final Assessment (viva and project)	**13, 14**

I. Assessment Tools:

Assessment Tools	**Weightage (%)**
1. Attendance	5 %
2. Assignment	5 %
3. Quiz	10 %
4. Lab works, project & viva	25 %
5. Midterm Examination	25 %
6. Final Examination	30 %
Total	**100 %**

J. CO Assessment Plan:

Assessment Tools	**Course Outcomes (CO)**				
	CO1	**CO2**	**CO3**	**CO4**	**CO5**
Attendance	-	-	-	-	-
Assignment	-	-	-	-	-
Quiz	-	-	-	-	-
Lab works, project & viva					√
Midterm Examination	√	√			
Final Examination		√	√	√	